



HR Analytics — Employee Attrition Analysis

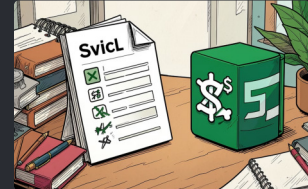
End-to-end project using Python, SQL, and Power BI to analyze employee attrition, identify drivers of turnover, and recommend actions to improve retention.

Project Snapshot



Tools

Python (Pandas, NumPy, Matplotlib, Seaborn), MySQL, Power BI, Excel, Google Colab.



Artifacts

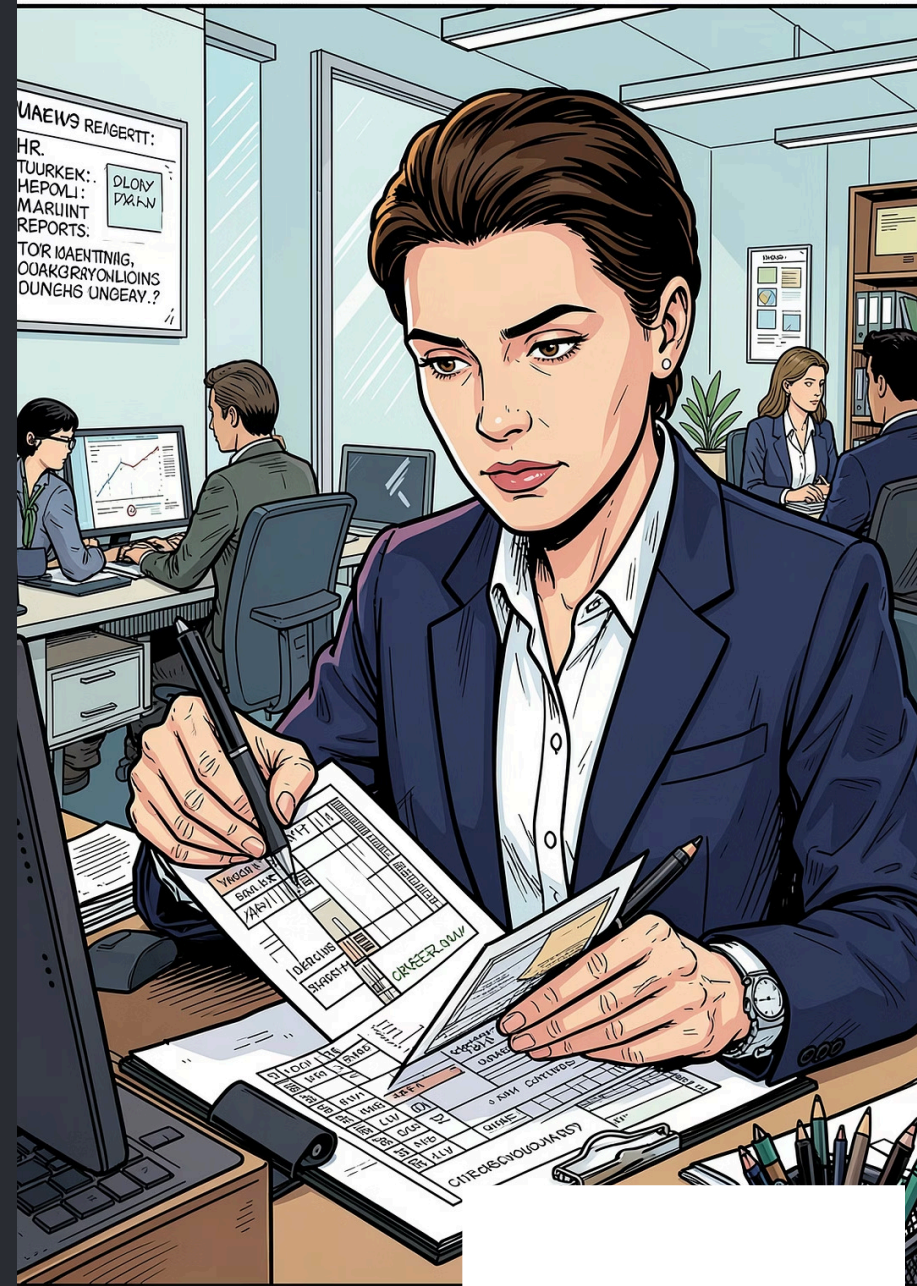
Notebook, SQL scripts, dataset CSV, Power BI dashboards, report documentation.



Problem Statement

Employee attrition increases recruitment costs and reduces productivity.

Objective: identify drivers of turnover and deliver data-driven recommendations to improve retention.



Project Workflow



This workflow structures the analysis from raw data to actionable insights and dashboards for stakeholders.



Key Analyses Performed



Overall Attrition

Aggregate attrition rate analysis and temporal trends.



Department-wise

Attrition breakdown by department (Sales highlighted).



Job Role Analysis

Turnover patterns across job roles.



Salary vs Attrition

Income levels correlated with likelihood to leave.



Overtime Impact

Employees working overtime show higher attrition.



Job Satisfaction

Low satisfaction strongly associated with turnover.



Dashboard & KPIs



Total Employees • Attrition Count • Attrition Rate

Average Monthly Income • Department-wise attrition • Job role analysis
• Demographics • Interactive filters & KPIs

Key Insights

Overtime

Employees working overtime have higher attrition rates.

Job Satisfaction

Low satisfaction strongly correlates with turnover.

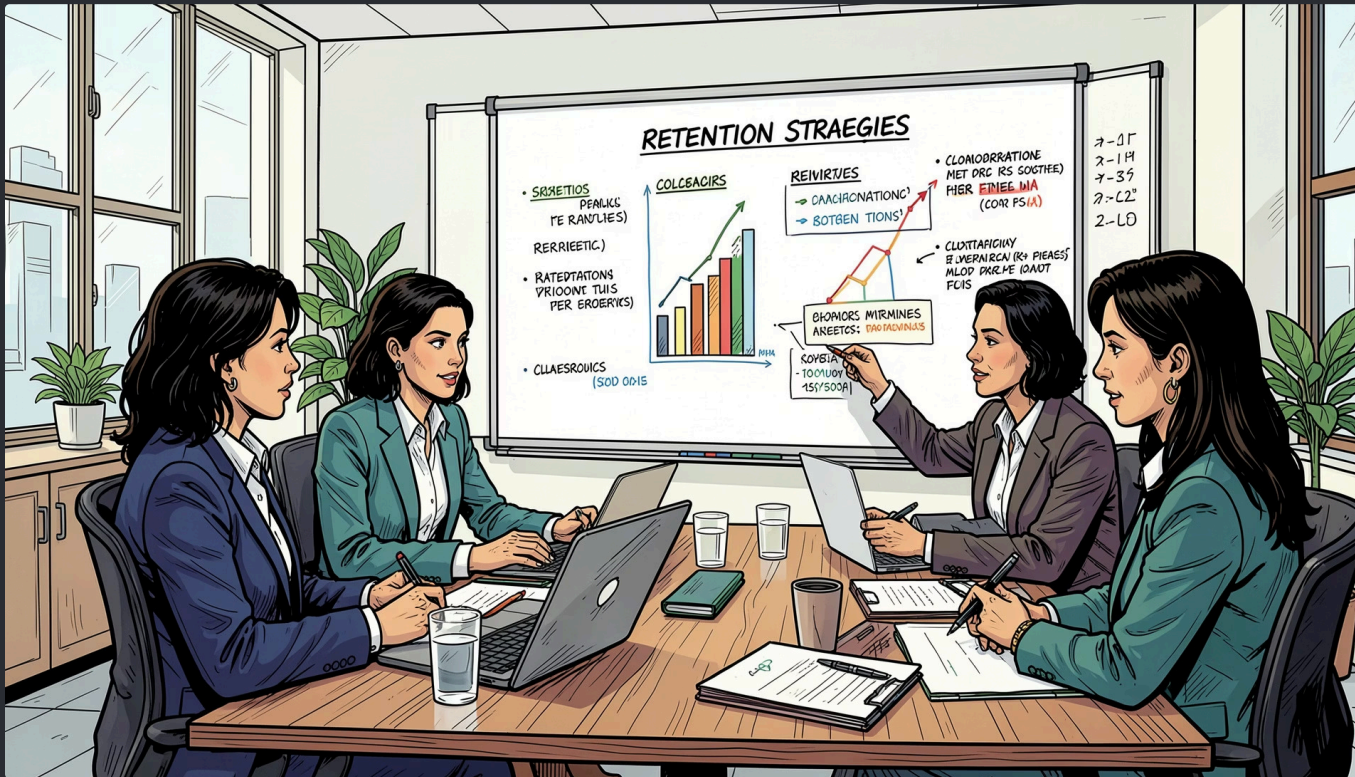
High-Risk Groups

Sales department shows the highest attrition; early-tenure employees leave most often.

Compensation

Lower income levels are associated with greater likelihood to leave.

Business Recommendations

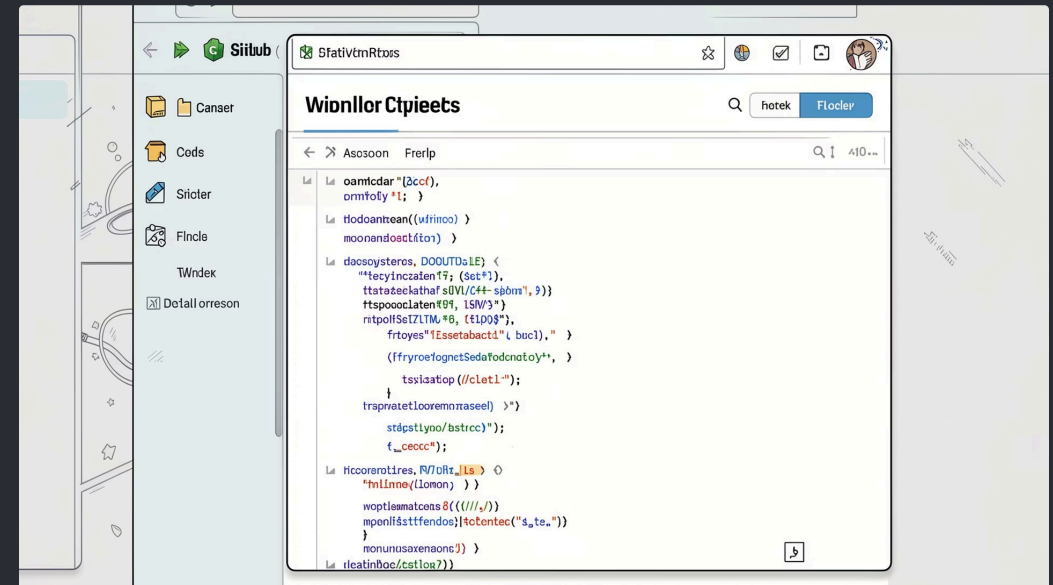


- Improve employee engagement programs.
- Reduce excessive overtime.
- Enhance career growth and learning opportunities.
- Review compensation and benefits structure.
- Conduct regular employee satisfaction surveys.

Repository & Resources

Repository includes:

- HR_Analytics_Analysis.ipynb
- SQL Analysis.docx
- python_sql_powerbi_project_dataset.csv
- Power BI dashboard files & screenshots



Author & Next Steps



Amit Kumar — Aspiring Data Analyst

Skills demonstrated: Data cleaning, EDA, SQL querying, visualization, dashboard development, business analytics.

- ❏ Next steps: Implement recommended retention programs, monitor KPIs in dashboard, iterate with updated data.